

Universal Simulation of Multiparticle Hamiltonians: Efficient Schemes and Complexity Bounds

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We address the problem of simulating pair-interaction Hamiltonians in n node quantum networks where the subsystems have arbitrary, possibly different, dimensions. We show that any pair-interaction can be used to simulate any other by applying sequences of appropriate local control operations. Efficient schemes for decoupling and time reversal can be constructed from orthogonal arrays. Conditions on time optimal simulation are formulated in terms of spectral majorization of matrices characterizing the coupling parameters.

Control-theoretical model of quantum computing

Given: **fixed pair-interaction Hamiltonian** acting on n nodes of dimension d

$$H = \sum_{1 \leq l < k \leq n} H_{kl}$$

External control: **local unitaries** on each node

$$V = w_1 \otimes w_2 \otimes \cdots \otimes w_n \in SU(d) \otimes \cdots \otimes SU(d)$$

Fast control limit: local unitaries can be performed arbitrarily fast

Continuous algorithm: sequence of time evolutions and local transformations

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

Complexity of a unitary is running time of the fastest algorithm

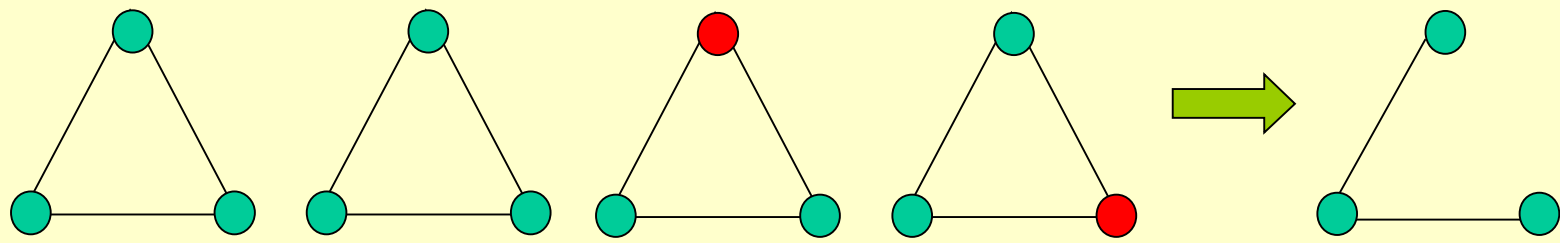
Complete zz-Hamiltonian

- defines quantum complexity without gates [2].
- illustrates the connection between the questions of optimal simulation of Hamiltonians and of separability of quantum states [1].

Use **complete** zz-Hamiltonian to simulate zz-Hamiltonians with arbitrary graphs

$$H := \sum_{k < l} \sigma_z^k \sigma_z^l \quad \tilde{H} := \sum_{k < l} a_{kl} \sigma_z^k \sigma_z^l$$

Conjugation of k -th spin by σ_x flips the sign of all interactions connected to k

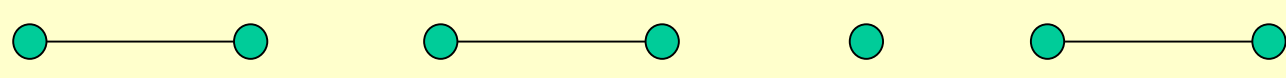


cancelling certain interactions requires time overhead
here 4 time periods of duration $\frac{1}{2}$ produce the desired graph (overhead 2)

Defining quantum complexity without gates

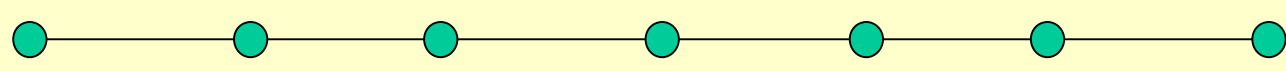
The complete graph complexity (CGC) of a unitary is defined as the minimal running time for a quantum computer with complete zz-Hamiltonian

- implements parallel two qubit gates with no overhead (chromatic index is 1)



(CGC is less than the weighted depth of a parallelized two-qubit network)

- simulates short-range interactions with constant overhead



(nearest neighbors in one dimension: overhead 2)

- allows higher parallelization than in a two-qubit network

Simulating Hamiltonian time evolutions

Let $\tilde{H}$ be any pair-interaction Hamiltonian. The time evolution $\exp(-i\tilde{H}t)$ according to $\tilde{H}$ is simulated by using $\exp(-iHt)$ for **arbitrary small time steps** and local operations. Only Hamiltonians in the linear span of local conjugates of $\tilde{H}$ are simulated (therefore no commutators are necessary).

The running time does not tend to infinity with increasing accuracy.

Def.: H can simulate $\tilde{H}$ with overhead 1 ($\tilde{H} \prec H$) iff $\tilde{H}$ is a convex combination of local conjugates of H . H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \prec \tau H$ (convex problem)

$$\tilde{H} = \sum_{j=1}^N p_j U_j^\dagger H U_j \Rightarrow e^{-i\tilde{H}\epsilon} \approx \prod_{j=1}^N U_j^\dagger e^{-iH p_j \epsilon} U_j$$

Simulating arbitrary pair-interactions by the complete zz-Hamiltonian

general Hamiltonian: $H_J := \sum_{kl; \alpha\beta} J_{kl; \alpha\beta} \sigma_\alpha^k \sigma_\beta^l$

J is a $3n \times 3n$ real symmetric matrix with $n \times 3 \times 3$ zero blocks on the diagonal.

Theorem [Optimality]: An interaction with matrix J can be implemented with no time overhead (by the complete zz-Hamiltonian) iff there is a separable n -qubit state ρ such that J is the $3n \times 3n$ -matrix

$$\left(\text{tr}(\rho \sigma_\alpha^k \sigma_\beta^l) \right)_{kl; \alpha\beta}$$

maximal strength of **pair-interaction** implementable in real time corresponds to maximal strength of **pair-correlation** in a separable state

Bounds on time overhead

- Lower bound: absolute value of the least eigenvalue of J [1]
- Upper bound: **weighted chromatic index** of the interaction graph [3]
(Generalization of chromatic index for weighted graphs; weights given by operator norms)

Simulation of Hamiltonians in quantum networks

The notion of the J -matrix generalizes to systems consisting of d -dimensional subsystems ($d > 2$) by choosing an orthonormal basis $B := \{\sigma_\alpha\}_{\alpha=1}^{d^2-1}$ of the Lie algebra $\mathfrak{su}(d)$.

$$H := \sum_{k < l} H_{kl} \quad J := \begin{pmatrix} 0 & J_{12} & J_{13} & \cdots & J_{1n} \\ J_{21} & 0 & J_{23} & \cdots & J_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ J_{n1} & J_{n2} & J_{n3} & \cdots & 0 \end{pmatrix}$$

$$H_{kl} := \sum_{\alpha\beta} J_{kl; \alpha\beta} \sigma_\alpha^k \sigma_\beta^l \quad J_{kl} := (J_{kl; \alpha\beta})_{\alpha, \beta}$$

$$H \mapsto (u \otimes v) H_{kl} (u \otimes v)^\dagger \quad J_{kl} \mapsto U J_{kl} V^T$$

Theorem [5]: Let H and $\tilde{H}$ be any pair-interaction Hamiltonians. A necessary condition that $\tilde{H}$ can be simulated with overhead τ by H is that **the spectrum of $\tilde{J}$ is majorized by the spectrum of τJ** . Furthermore, it is necessary that this criterion is still satisfied after rescaling the couplings between the nodes.

Decoupling and inversion in quantum networks

Efficient **decoupling schemes** for networks can be constructed with the help of **orthogonal arrays**. The entries of the orthogonal array determine which local operations of an annihilator have to be applied on the nodes [5].

The decoupling schemes give a method to achieve **universal simulation** [5].

The problem to invert a Hamiltonian H is to simulate $-H$. There are physical Hamiltonians such that the overhead for inversion grows linearly with the number of nodes (Arrow of time in quantum control) [2].

A normal form for orthogonal arrays can be used to construct efficient **inversion schemes** [5].

- [1] P. Wocjan, D. Janzing, and Th. Beth, *Simulating arbitrary pair-interactions by a given Hamiltonian: Graph-theoretical Bounds on the Time Complexity*, quant-ph/0106077.
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- [3] D. Janzing, Th. Beth, *Complexity measure for continuous time quantum algorithms*, quant-ph/0009094, Phys. Rev. A, vol. 64, no. 2, pp. 022301-1-6.
- [4] P. Wocjan, M. Rötteler, D. Janzing, and Th. Beth, *Universal simulation of Hamiltonians using a finite set of control operations*, quant-ph/0109063.
- [5] P. Wocjan, M. Rötteler, D. Janzing, and Th. Beth, *Simulating Hamiltonians in quantum networks: efficient schemes and complexity bounds*, quant-ph/0109088.